

Strong duality

Entropic Dual

- Dual Formulation:

$$\max_{\mathbf{f} \in \mathbb{R}^n, \mathbf{g} \in \mathbb{R}^m} \langle \mathbf{f}, \mathbf{a} \rangle + \langle \mathbf{g}, \mathbf{b} \rangle - \varepsilon \langle e^{\mathbf{f}/\varepsilon}, \mathbf{K} e^{\mathbf{g}/\varepsilon} \rangle \quad \left(= \langle f, \alpha \rangle + \langle g, \beta \rangle - \varepsilon \sum_{i,j} \exp\left(\frac{f_i + g_j - C_{ij}}{\varepsilon}\right) \right)$$

- **Strong duality** [eg P&C Prop 4.4]: entropic dual $\equiv$ entropic primal
- The optimal $\mathbf{f}, \mathbf{g}$ are related to the Sinkhorn scaling vectors $\mathbf{u}, \mathbf{v}$ as:
 $(\mathbf{u}, \mathbf{v}) = (e^{\mathbf{f}/\varepsilon}, e^{\mathbf{g}/\varepsilon})$
- This + known form of the coupling gives us a recipe to go from primal to dual sol:
 $\pi^\star = \text{Diag}(u) K \text{Diag}(v) \quad \Longleftrightarrow \quad \pi_{ij}^\star = \exp\left(\frac{f_i + g_j - C_{ij}}{\varepsilon}\right).$

Entropic Dual: General Case

$$\sup_{(f,g) \in \mathcal{C}(\mathcal{X}) \times \mathcal{C}(\mathcal{Y})} \int_{\mathcal{X}} f d\alpha + \int_{\mathcal{Y}} g d\beta - \varepsilon \int_{\mathcal{X} \times \mathcal{Y}} e^{\frac{-c(x,y) + f(x) + g(y)}{\varepsilon}} d\alpha(x) d\beta(y) .$$